

**McGill University**  
**ECSE 222 – Digital Logic (Fall 2025)**  
**Pen-and-Paper Assignment #6**

**Instructions**

- TA in charge: Sihui Shen (sihui.shen@mail.mcgill.ca)
- Write/type your answers in the space provided.
- Due date: Monday, Dec 8<sup>th</sup>, 2025, by 11:59 p.m. EDT.
- You must electronically submit the assignment in the form of a PDF file (only this format will be accepted) on myCourses before the deadline. You may submit as many times as you want, however, only the last submission will be kept by the system.
- You must solve the assignment in teams of two students, whereas only one of the students should perform the actual electronic submission.
- Late submissions will incur penalties as described in the course syllabus.

**Student identification**

|                                         |  |
|-----------------------------------------|--|
| First student full name:                |  |
| First student ID number:                |  |
| Second student full name:               |  |
| Second student ID number:               |  |
| Third student full name (if applicable) |  |
| Third student ID number (if applicable) |  |

**Evaluation (for course graders use only)**

| Question 1<br>(20%) | Question 2<br>(10%) | Question 3<br>(20%) | Question 4<br>(20%) | Question 5<br>(15%) | Question 6<br>(15%) | <b>Total</b> |
|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|--------------|
|                     |                     |                     |                     |                     |                     |              |

**Question 1 (20%)**

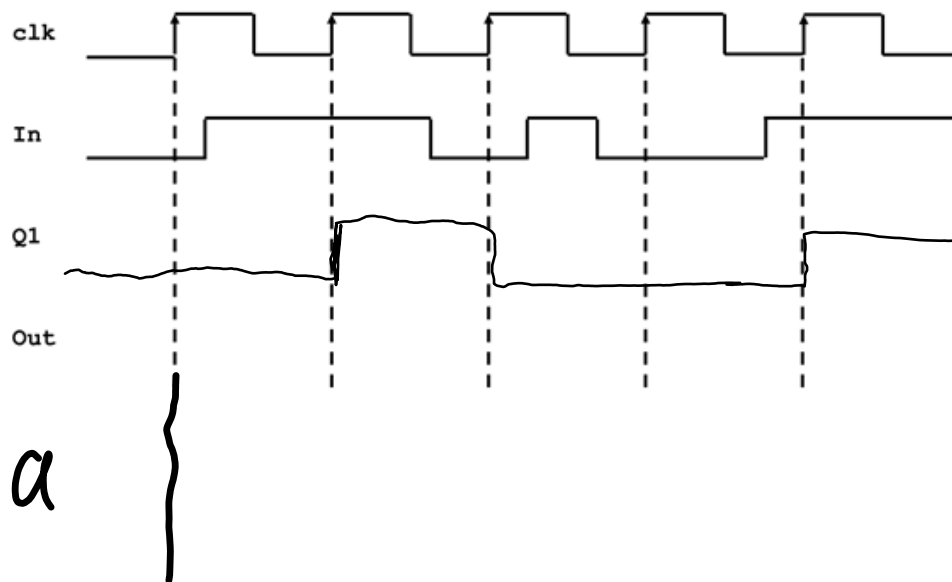
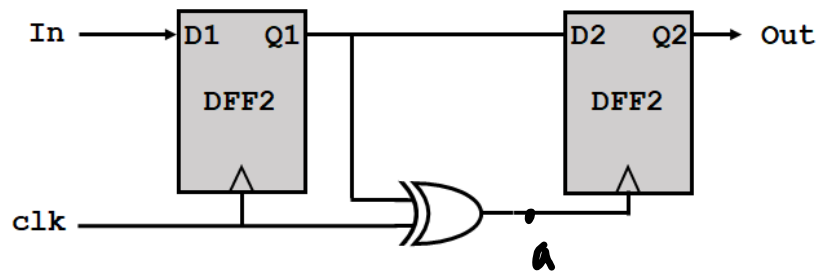
Design the circuit of a new funky full of choices FF called CFF. The CFF has four inputs:  $D$ ,  $\bar{L}/FF$ ,  $\bar{N}/P$ , and  $clk$ , and one output:  $Q$ . The CFF operates as follows:

- $D$  is the data input
- $clk$  is the clock input
- The CFF can operate either as a latch or as a FF. The input  $\bar{L}/FF$  chooses between the two modes of operation ('0' for latch and '1' for FF).
- The CFF can operate at the positive or negative level/edge (level for latch and edge for FF) of the clock. The input  $\bar{N}/P$  chooses between the two modes of operation ('0' for negative and '1' for positive).

You may use standard sequential elements (as shown in class) without providing their internal circuitry, and standard logic gates (as few as possible) with at most two inputs (no MUXs allowed).

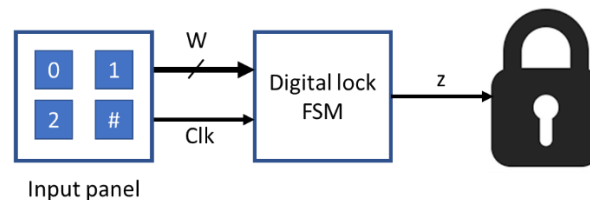
**Question 2 (10%)**

For the following circuit, complete the timing diagram below. Provide your answer on the given timing diagram. Assume ideal components (no latency). Initial state  $Q1Q2 = "00"$ .



**Question 3 (20%)**

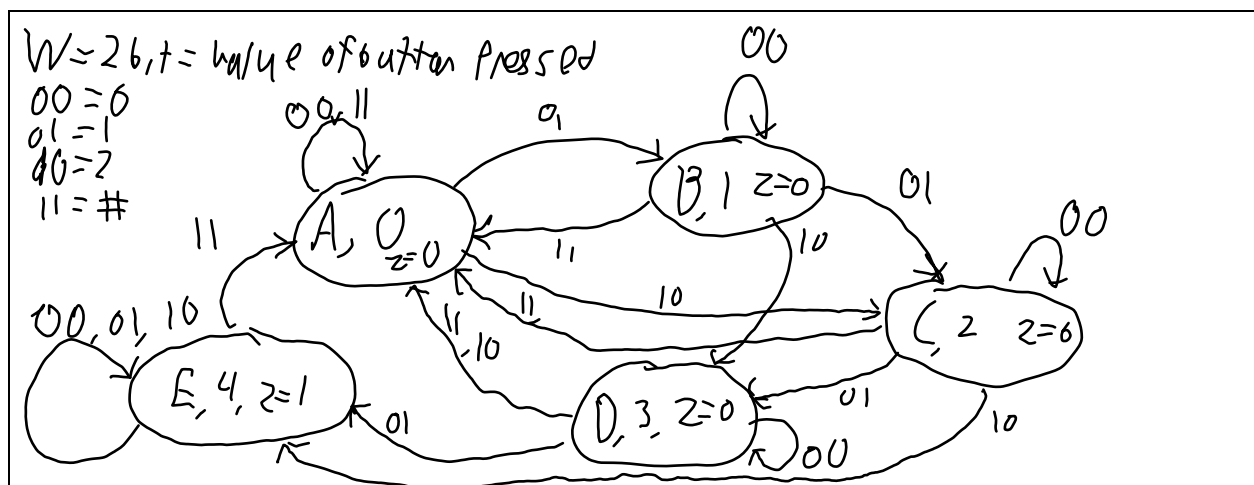
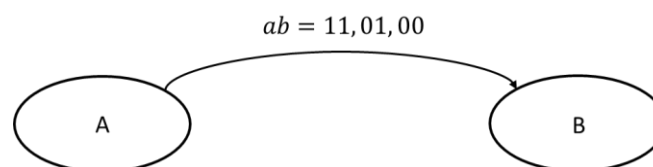
Design a digital lock system. The system includes three components (as depicted in the figure below): (1) an input panel that consists of four buttons, (2) an FSM that receives the punched-in code 'W' in binary form, and generates an output signal 'z', and (3) a lock that unlocks when  $z = 1$ .



Specifications:

- When any button is pressed on the input panel, a binary sequence 'W' is generated and sent to the FSM. At the same time, a single clock pulse (rise and fall) is also generated and sent to the FSM, assume that all timing constraints are met.
- The digital lock includes a summation unit such that the lock only opens when the sum of all pressed numbers is 4, producing an output  $z = 1$ , otherwise  $z = 0$ . If the value in the summation unit exceeds 4, it automatically resets to zero.
- The '#' button is used to lock the system and reset the summation unit in the FSM to zero.
- Assume that the initial state of the system is in the locked condition and the value in the summation unit is zero.
- Assume that once unlocked, the system stays unlocked until the '#' button is pressed.

Design and implement the digital lock FSM as a Moore machine using DFFs. You must determine the size of 'W' and provide all the steps of your solution. Note that if two different input combinations point from one state to the same next state, they may be combined on a single edge as illustrated here:



**State table**

| Present State | W=00 | W=01 | W=10 | W=11 | Z |
|---------------|------|------|------|------|---|
| A             | A    | B    | C    | A    | 0 |
| B             | B    | C    | D    | A    | 0 |
| C             | C    | D    | E    | A    | 0 |
| D             | D    | E    | A    | A    | 0 |
| E             | E    | E    | E    | A    | 1 |

**State assigned table**

| $Y_2, Y_1, Y_0$ | 00  | 01  | 10  | 11  | Z |
|-----------------|-----|-----|-----|-----|---|
| 000             | 000 | 001 | 010 | 000 | 0 |
| 001             | 001 | 010 | 011 | 000 | 0 |
| 010             | 010 | 011 | 100 | 000 | 0 |
| 011             | 011 | 100 | 000 | 000 | 0 |
| 100             | 100 | 100 | 100 | 000 | 1 |

**Next State Karnaugh Maps**

$Y_0$  K-map:

| $Y_1, Y_0$ | 00 | 01 | 11 | 10 |
|------------|----|----|----|----|
| 00         | 0  | 1  | 1  | 0  |
| 01         | 1  | 0  | 0  | 1  |
| 11         | 0  | 0  | 0  | 0  |
| 10         | 0  | 1  | 0  | 0  |

$Y_1$  K-map:

| $Y_1, Y_0$ | 00 | 01 | 11 | 10 |
|------------|----|----|----|----|
| 00         | 0  | 0  | 0  | 0  |
| 01         | 0  | 0  | 0  | 0  |
| 11         | 0  | 0  | 0  | 0  |
| 10         | 0  | 0  | 0  | 0  |

**Output Equations**

$Z = Y_2$

$Y_0 = Y_1 \bar{W}_1 \bar{W}_0 + Y_1 \bar{Y}_0 \bar{W}_1 W_0 + \bar{Y}_1 Y_0 W_1 \bar{W}_0 + \bar{Y}_1 Y_0 \bar{W}_1 W_0 \bar{Y}_2$

$Y_1$  K-map:

| $Y_1, Y_0$ | 00 | 01 | 11 | 10 |
|------------|----|----|----|----|
| 00         | 0  | 0  | 1  | 1  |
| 01         | 0  | 1  | 0  | 1  |
| 11         | 0  | 0  | 0  | 0  |
| 10         | 1  | 1  | 0  | 0  |

$Y_1 = Y_1 \bar{Y}_0 \bar{W}_1 + Y_1 \bar{W}_1 \bar{W}_0 \bar{Y}_2 + \bar{Y}_1 Y_0 \bar{W}_1 W_0 + \bar{Y}_1 W_1 \bar{W}_0 \bar{Y}_2$

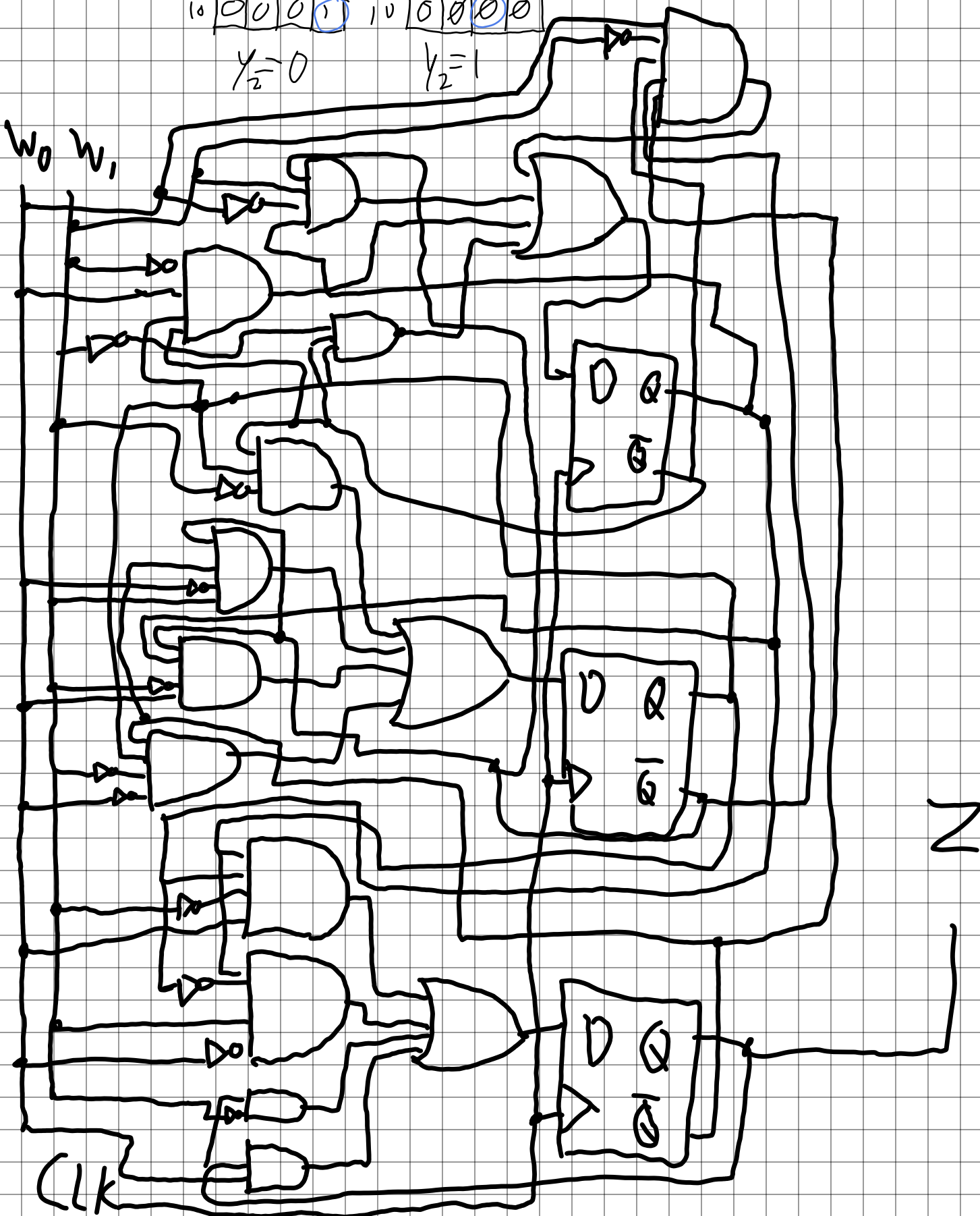
Y  
2

|   |    | $y_1, y_0$ |   |    |    |    |    | $x_1, x_0$ |   |            |   |    |    |    |    |
|---|----|------------|---|----|----|----|----|------------|---|------------|---|----|----|----|----|
|   |    | $w_1, w_0$ |   | 00 | 01 | 11 | 10 |            |   | $w_1, w_0$ |   | 00 | 01 | 11 | 10 |
| Y | 00 | 0          | 0 | 0  | 0  | 0  | 0  | 00         | 1 | 0          | 0 | 0  | 0  | 0  | 0  |
|   | 01 | 0          | 0 | 1  | 0  | 0  | 0  | 01         | 1 | 0          | 0 | 0  | 0  | 0  | 0  |
|   | 11 | 0          | 0 | 0  | 0  | 0  | 0  | 11         | 1 | 0          | 0 | 0  | 0  | 0  | 0  |
|   | 10 | 0          | 0 | 0  | 1  | 0  | 0  | 10         | 0 | 0          | 0 | 0  | 0  | 0  | 0  |

$y_2 = 0$

$y_2 = 1$

$$Y_2 = y_1 y_0 \bar{w}_1 w_0 + y_1 \bar{y}_0 w_1 \bar{w}_0 + y_2 \bar{w}_1 + y_2 w_0$$



**Question 4 (20%)**

A non-overlapping sequence detector is a machine that resets after detecting the desired sequence. Generate a state diagram (clearly explain the meaning of each state), relevant tables, and circuit for a non-overlapping “1001” sequence detector with the following specifications:

- Machine has a two-bit input  $\{b_1b_0\}$ : every clock cycle two bits of the sequence arrive
- MSB of the sequence should be detected first
- This machine has two outputs  $z_1z_0$
- Machine will output  $z_0 = 1$  as soon as the sequence is detected (in the same clock cycle)
- Machine will output  $z_1 = 1$  as soon as the sequence is detected twice in a row (in the same clock cycle)

Use as few Toggle FFs (TFFs) as possible and logic gates as needed.





**Question 5 (15%)**

A company named “Not Fun to Work At” is interested to install a sensor system at their Montreal office to monitor the movements of the two employees in that office. The sensor will always report the correct number of employees in the office (for all possible inputs). Design a Mealy FSM (using DFFs) with the following specifications (provide all relevant diagrams, tables, and circuits, also clearly identify the meaning of each state):

- The sensor system recognizes radio frequency (RF) tags embedded within employee IDs and thus will only identify employees (not other people)
- Every time an RF tag is identified, the system is also able to tell the direction of the movement, either into the office or outside
- Every time an RF tag is identified, a short clock pulse is generated within the system
- More than one employee may move through the door of the office at the same time (only in the same direction), the system is able to accommodate for this
- Assume that all timing constraints are met and that all employees carry their IDs at all times



**Question 6 (15%)**

As shown in class, one error detection technique is the addition of a parity bit to every transmitted word. Design a Mealy FSM (using DFFs) that receives four-bit words that include three data bits and one parity bit (a single bit arrives every clock cycle) and produces  $z = 1$  if an error is detected in the word. For every received word, the three data bits arrive first, followed by the parity bit. When the parity bit arrives, the system resets to the initial state. Assume that at most one error can occur. Assume that the output is only observed at the clock cycle during which the parity bit arrives. Hint: include an initial/reset state that means “Awaiting first bit of word.” For the FSM, provide all relevant diagrams, tables, and circuits. Also clearly identify the meaning of each state.

